

Hyper-parameters : search / optimization

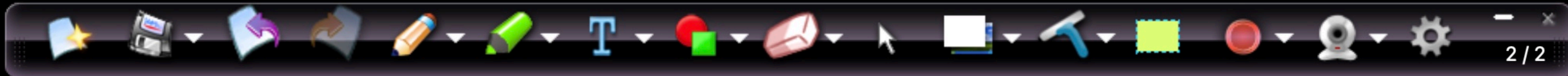
LR $\rightarrow \lambda \rightarrow ?$
KNN $\rightarrow K ?$
L₂ Reg $\rightarrow \lambda \rightarrow ?$

$\lambda \rightarrow$
 $\lambda = 0 \rightarrow \text{overfitting}$
 $\lambda = \infty \rightarrow \text{underfitting}$

Best parameters $\uparrow$

Q: How to determine the best λ $\rightarrow$ small data
Big data $\rightarrow$ Gridsearch / Randomsearch $\rightarrow$ sklearn
 $\rightarrow$ Bayesian optimization $\rightarrow$ 'Hyper opt'

eg: $\lambda \rightarrow \text{LA}(\lambda=0) \rightarrow \lambda = [0.001, 0.01, 0.1, 1, 10, 100, 1000]$



parameter

find λ $\rightarrow$ why $\rightarrow$ min error

Best λ $\rightarrow$ $\lambda = 100$, $\lambda = 1000$

$LR(\lambda_1 = 0.01, \lambda_2 = 0.1)$

eq 2

Here I have 2 Hyper parameters $\rightarrow$

λ_1

$M1 = LR(\lambda = 0.01)$

$\rightarrow M1.predict(X_{dev})$

$\rightarrow$ score $\rightarrow$ error

$\downarrow$

$M2 = LR(\lambda = 0.1)$

$M2.predict(X_{dev})$

$\rightarrow$

$\rightarrow$ $P_{train} 64\%$

$\rightarrow$ $P_{dev} 16\%$

$\rightarrow$ P_{test}

error



$$\lambda_2 = [0.001, 0.01, 0.1, 1, 10, 100, 1000]$$

time com

$\downarrow$

$$\lambda_1, \lambda_2 \rightarrow (m_1 \times m_2)$$

$$\hookrightarrow m^2$$

$\downarrow$

'K' $\rightarrow$ parameter

m-combinations

$$m^K$$

conclusion $\rightarrow$ as # hyper-parameters increases, the # of fine model need to the minimal increment experiment, why

$$m \rightarrow m^K$$

$$\rightarrow LR(\lambda_1 = 0.001, \lambda_2 = 0.001)$$

$\hookrightarrow$ error $\rightarrow$ $\boxed{X_1}$

$$LR(\lambda_1 = 0.001, \lambda_2 = 0.01)$$

$\hookrightarrow$ error $\rightarrow$ $\boxed{X_2}$

$$LR(\lambda_1 = 0.001, \lambda_2 = 0.1)$$

$\hookrightarrow$ error

$\Rightarrow$ Gridsearch is not working and here $\rightarrow$ memory $\rightarrow$ time $\downarrow$

1M $\rightarrow$ time $\rightarrow$ memory $\rightarrow$ 'GS'

'56ky' $\rightarrow$

Random Search (✓)

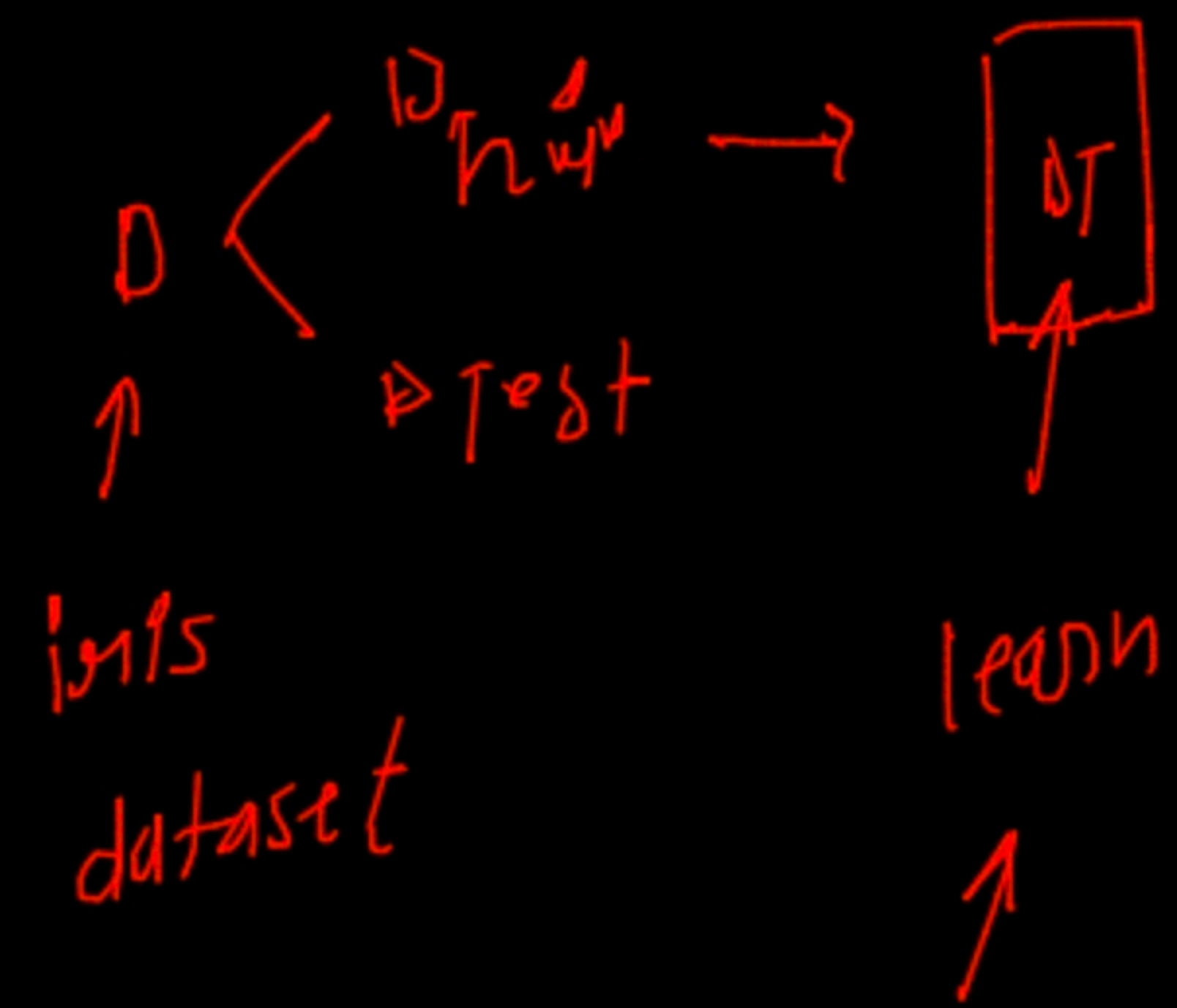
$$\lambda_1 = [0.001, 0.01, 0.1, 1]$$
$$\lambda_2 = [0.001, 0.01, 0.1, 1]$$

DT (Decision Tree) $\rightarrow$ if - else

- $\hookrightarrow$ KNN $\rightarrow$ distance based method
- $\hookrightarrow$ logReg $\rightarrow$ ~~prob~~ $\Rightarrow$ $\Rightarrow$
- $\hookrightarrow$ ~~linear~~ $\Rightarrow$

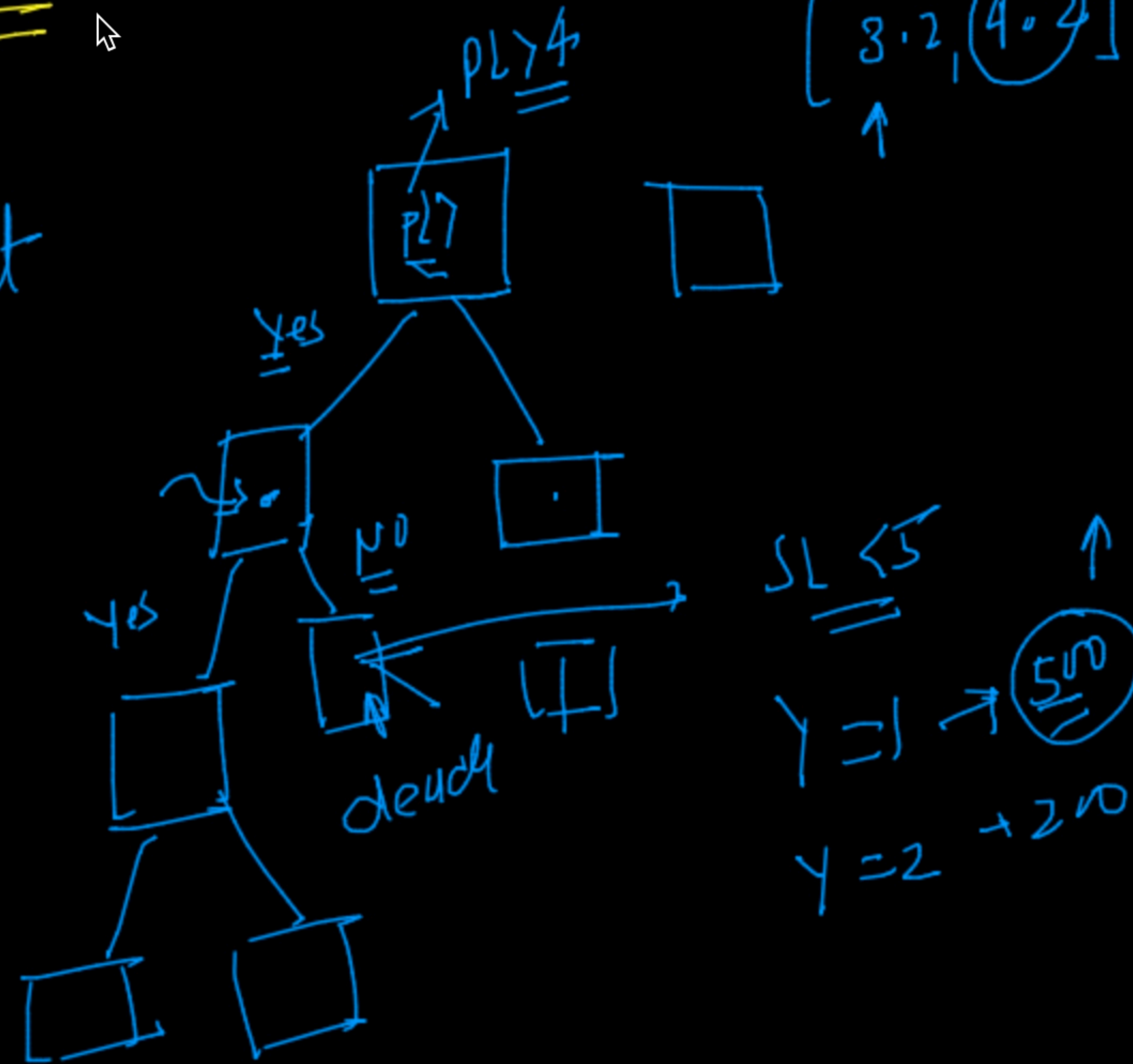
Nested if --- else classification / regression

eg,

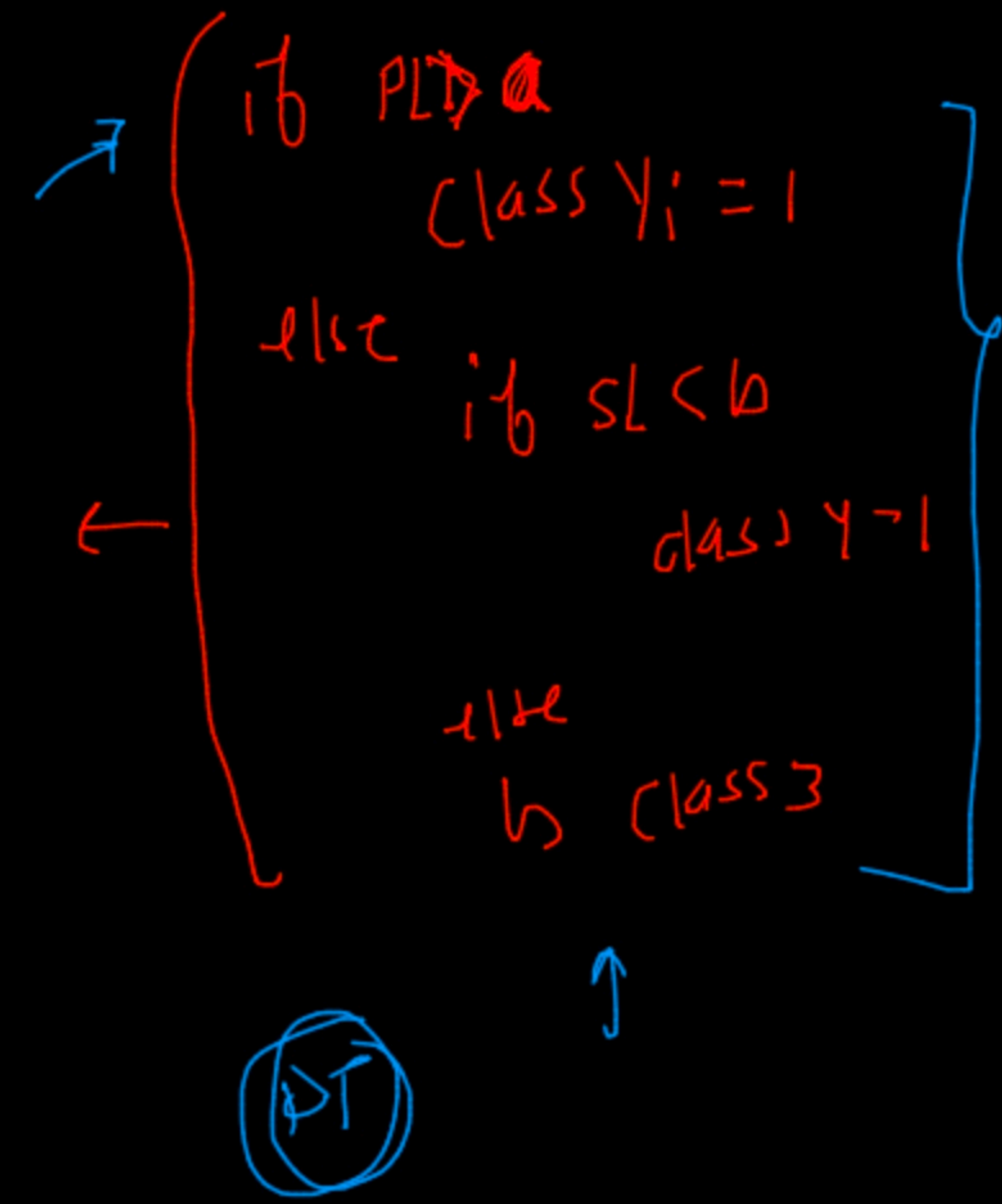


target

	1
	0

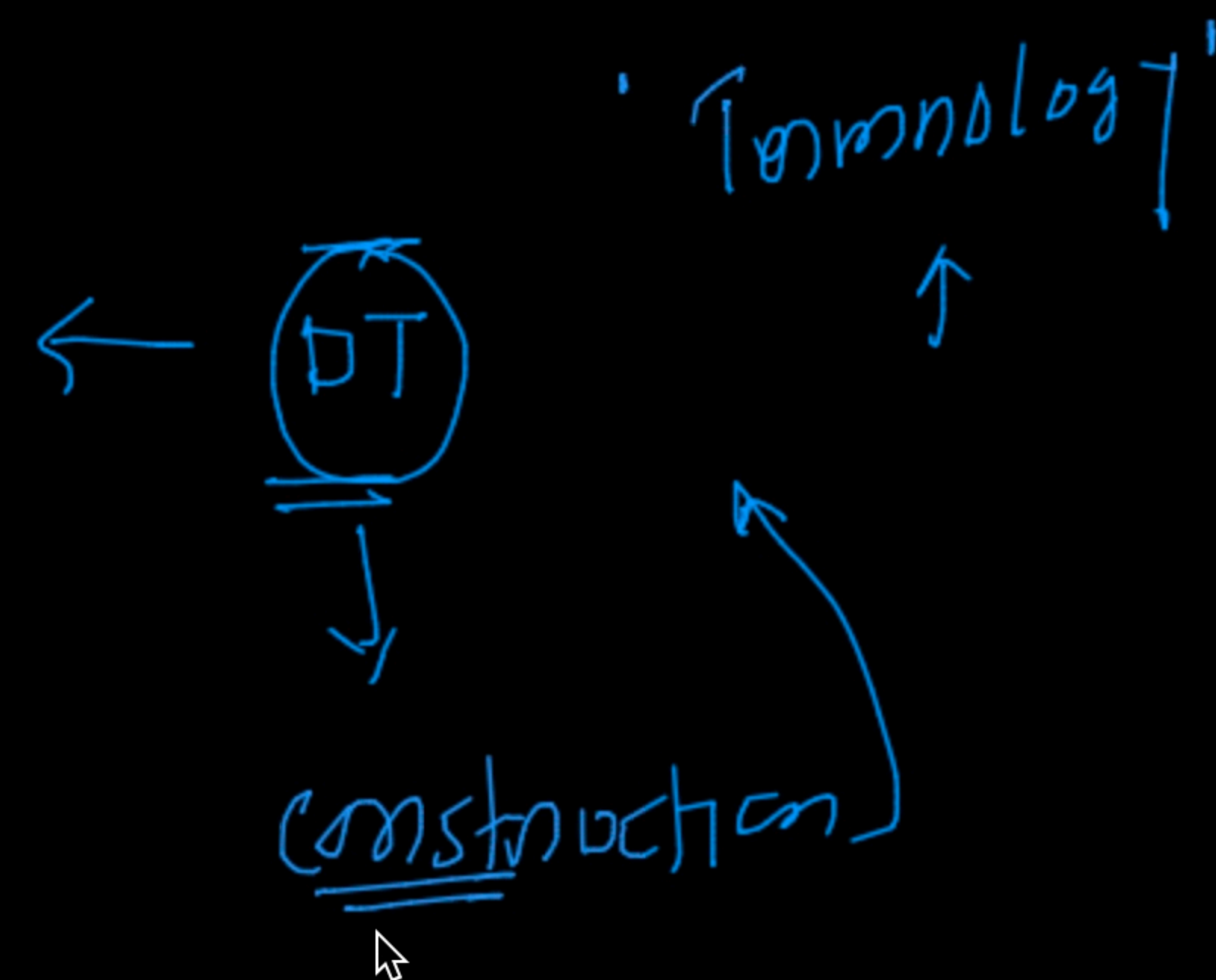
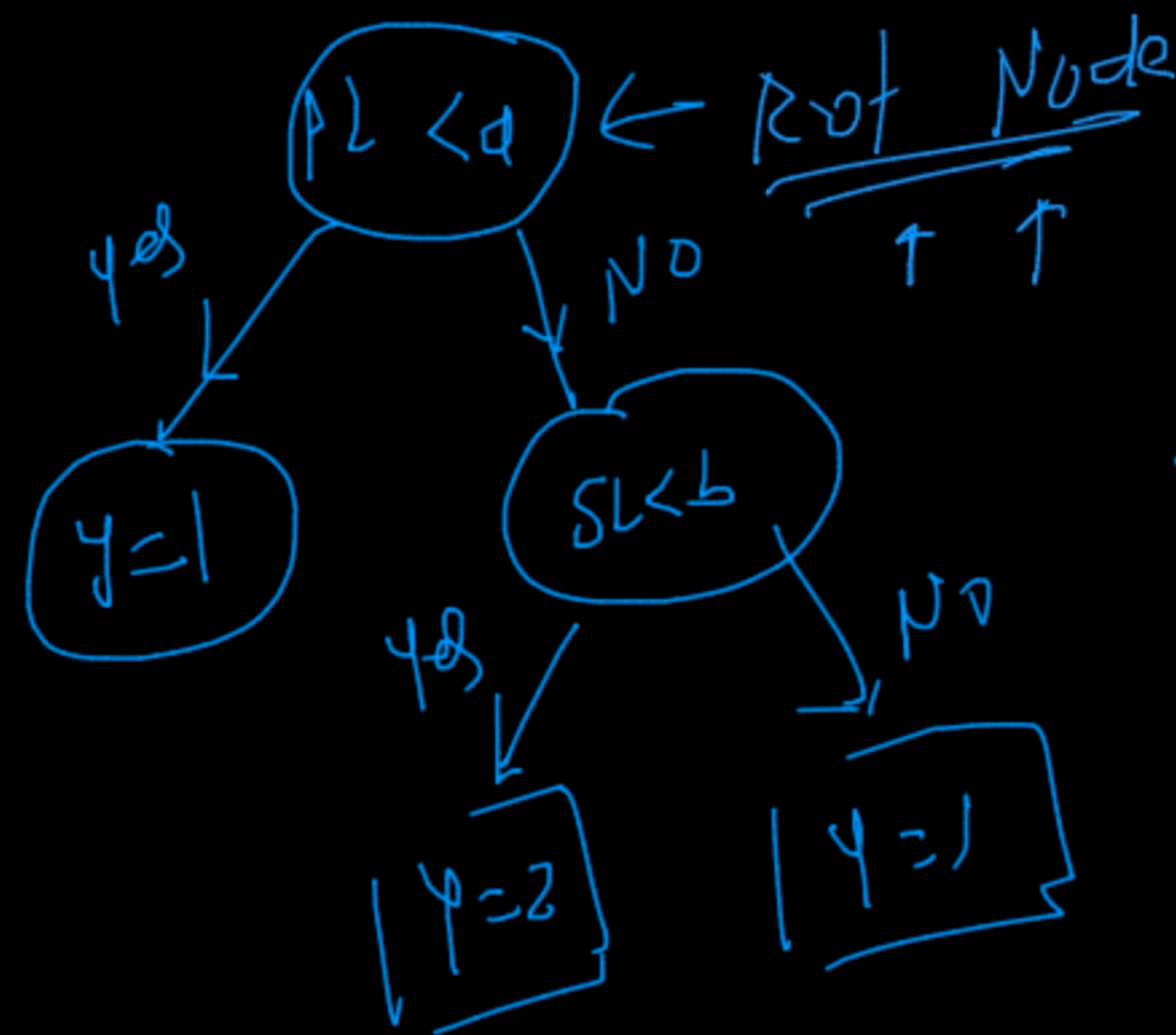


converted into



D → 4 fractions
x ↓

$[SL, PL, SW, PW]$
↑
↓
Criteria



1] ENTROPY → measure the randomness in your data → (Y)

↑

(Y)

Y → $x_1, x_2, x_3, \dots, x_k$

$$H(Y) = - \sum_{i=1}^k p(y_i) \cdot \log_b(p(y_i))$$

↑ $i=1$

↑ $b=2 \text{ or } e=2.718$

✓ $\left. \begin{array}{l} \text{No} \rightarrow 5 \\ \text{yes} \rightarrow 9 \end{array} \right\}$
14

All → $\left| \begin{array}{cc} 0 & 5 \\ 1 & 9 \end{array} \right|$
Per

→ 50/14

Y → $x_1, x_2, \dots, x_k$

play tennis →

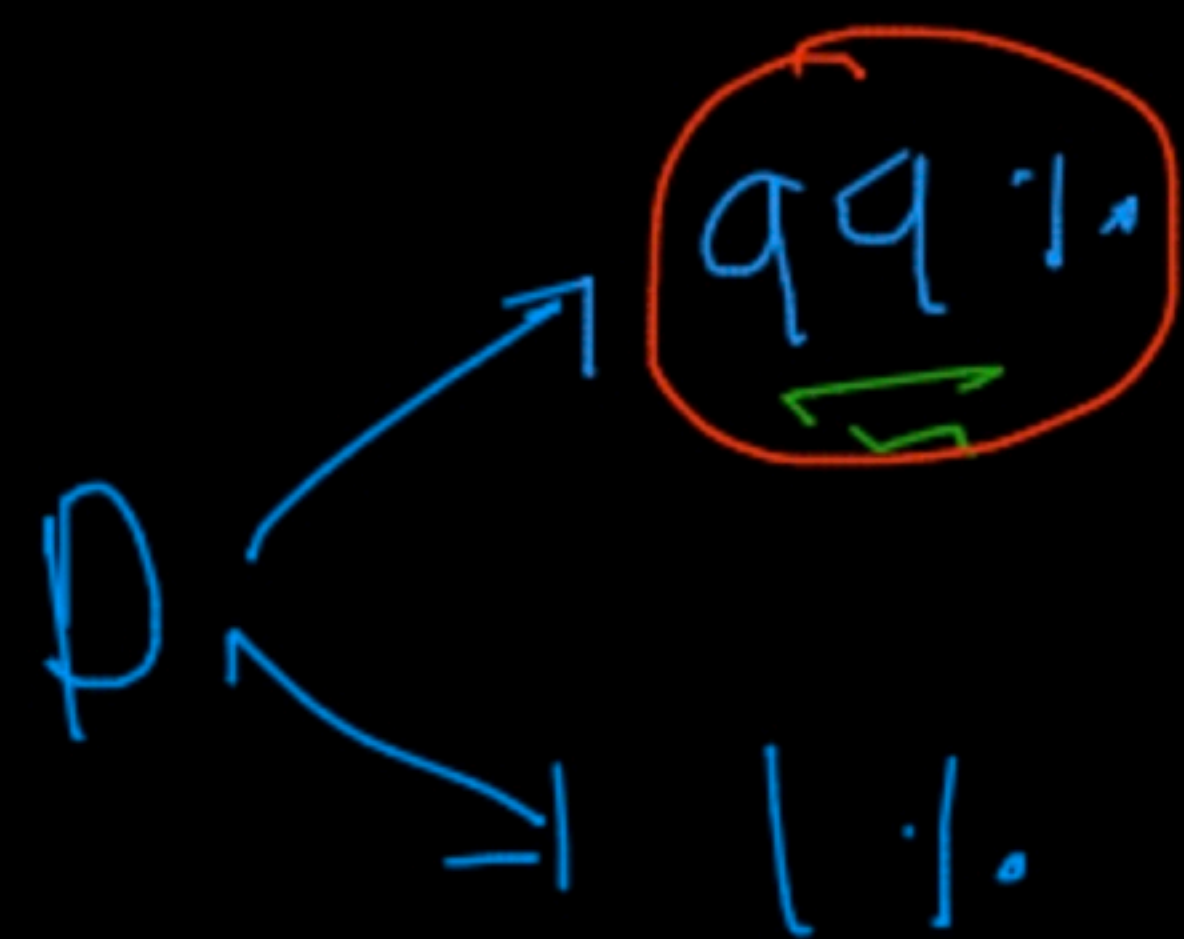
$$H(Y) = - \frac{9}{14} \log\left(\frac{9}{14}\right) - \frac{5}{14} \log\left(\frac{5}{14}\right)$$

$\uparrow$ $p_{Y=\text{No}}$ $\uparrow$ $p_{Y=\text{Yes}}$

→ $H(Y) \rightarrow \dots$

$Y \rightarrow Y_+ Y_-$ (2 class, 2 category)

Case i

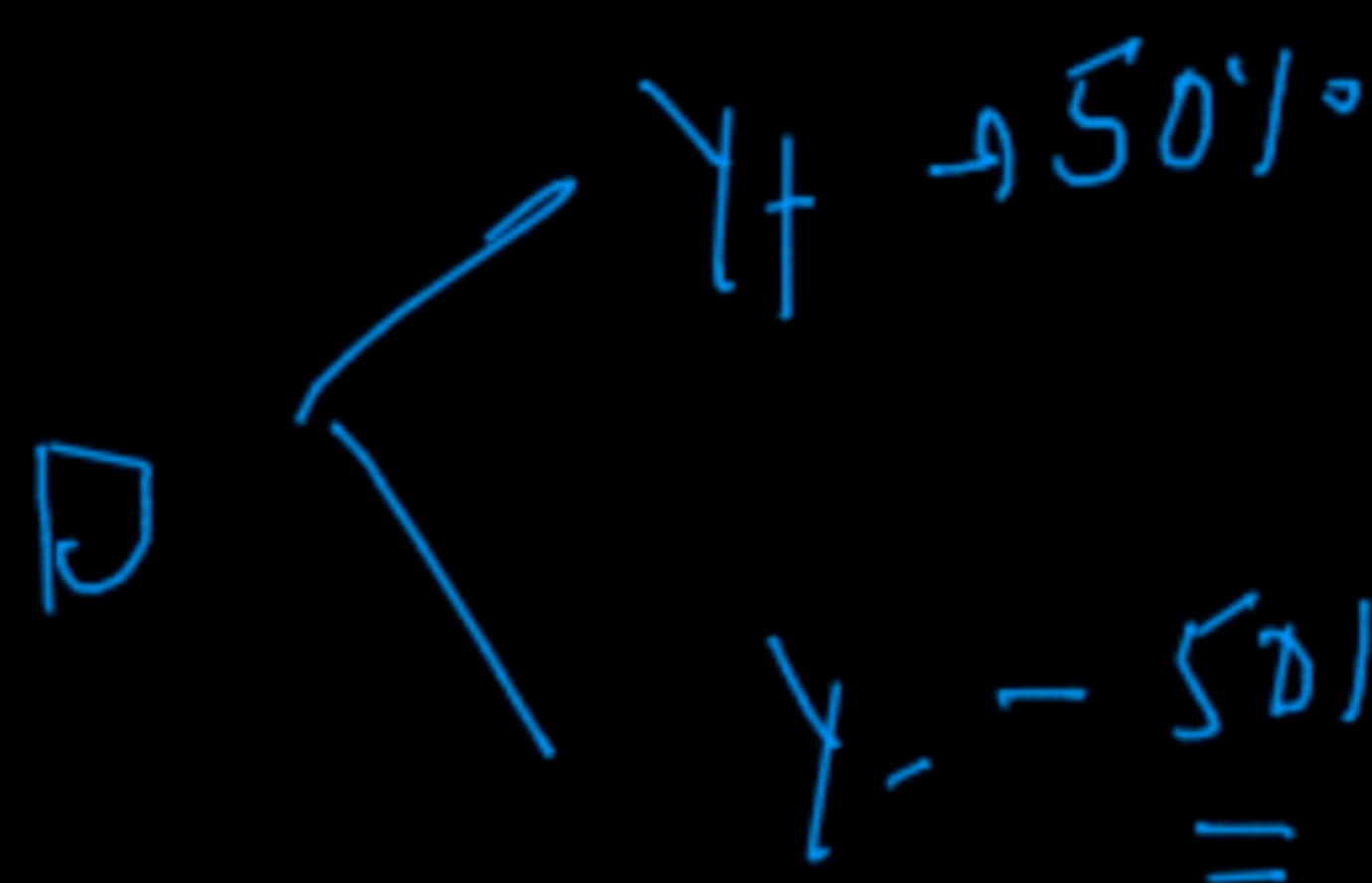


$$H(Y) = -0.99 \log(0.99) - 0.01 \log(0.01)$$

$\Downarrow$

$$= \underline{\underline{0.0801}} \rightarrow \underline{\underline{\text{small}}}$$

(Case ii)

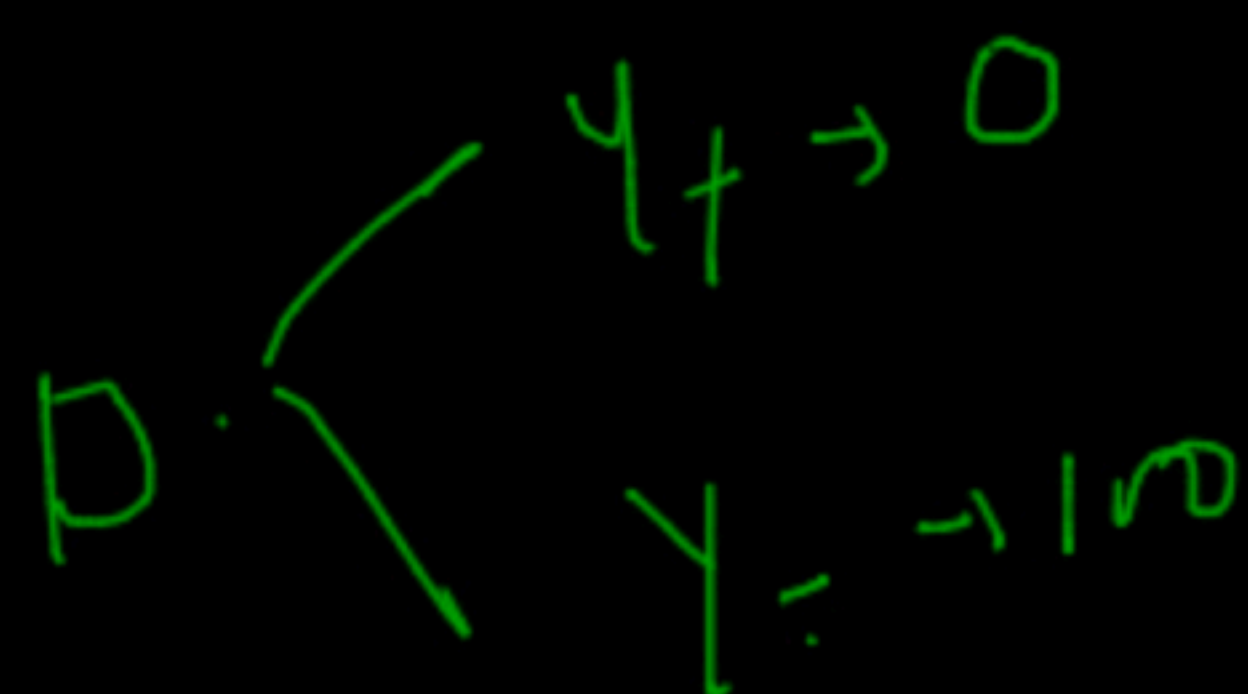


$$H(Y) = -0.5 \log(0.5) - 0.5 \log(0.5)$$

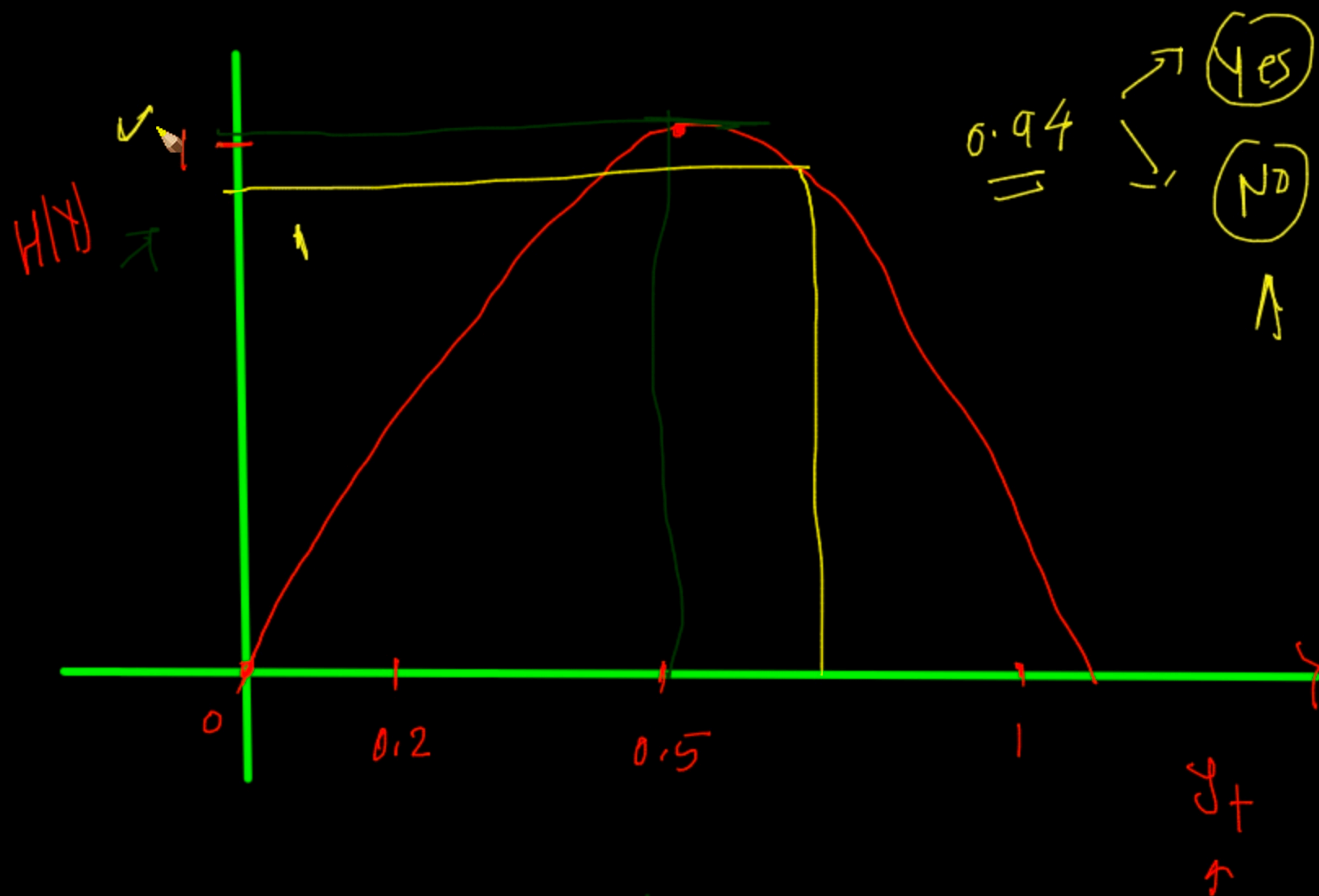
$\Downarrow$

$$= 1 \checkmark$$

(Case iii)



$$\rightarrow \boxed{H(Y) = 0}$$

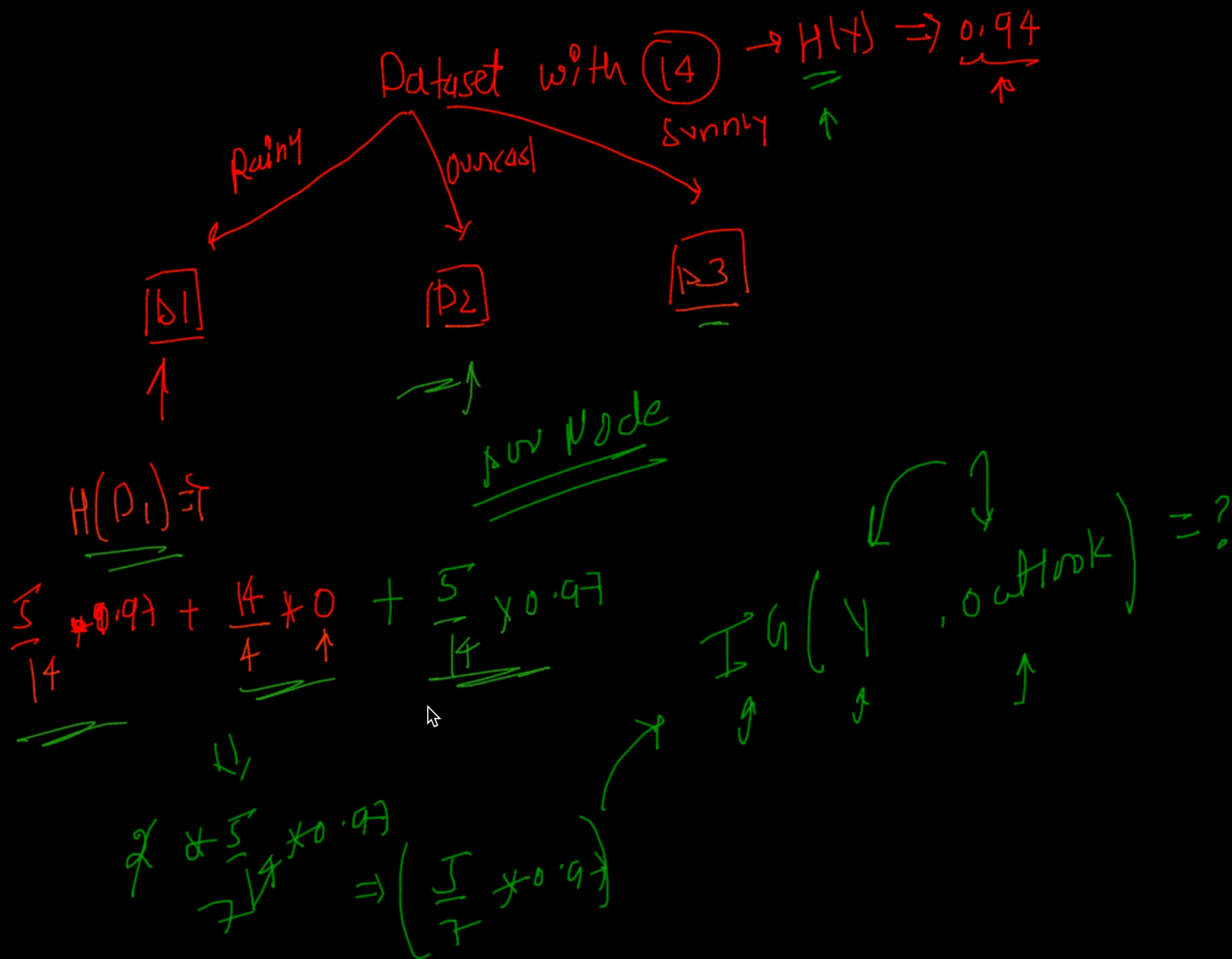


0.5

$(1-y^+)$

$\begin{cases} y^+ \rightarrow 0 \\ y^- \rightarrow 1 \end{cases} H/Y$

→ Information Gain (IG)

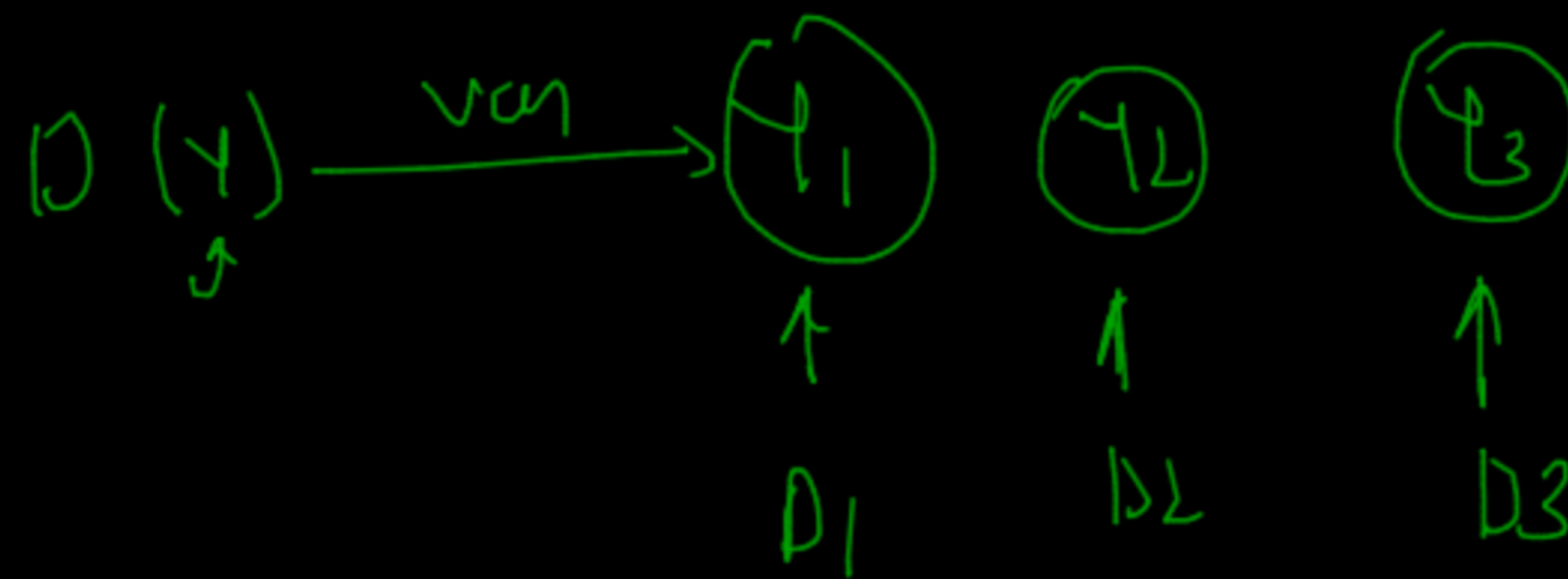


Why we call it Information gain?
Because we gain some information by breaking dataset into (D1 D2 D3)

$$IG(Y, outlook) = \left(\begin{array}{c} \text{weighted} \\ \text{entropy} \\ \text{after} \\ \text{breaking} \\ \text{into } D1 \\ D2 \\ D3 \end{array} \right) - \left(\begin{array}{c} \text{Entropy} \\ \text{before} \\ \text{breaking} \\ \text{the data} \end{array} \right)$$

$$= \frac{5}{7} * 0.97 - 0.94 \rightarrow$$

we can conclude as follows



Note $\rightarrow$ If value
lies b/w (0 to 1)

$$I_a(Y, \text{variable}) = \sum_{i=1}^k \frac{|D_i|}{|D|} * H(Y) - H(D|Y)$$

weighted entropy
after breaking